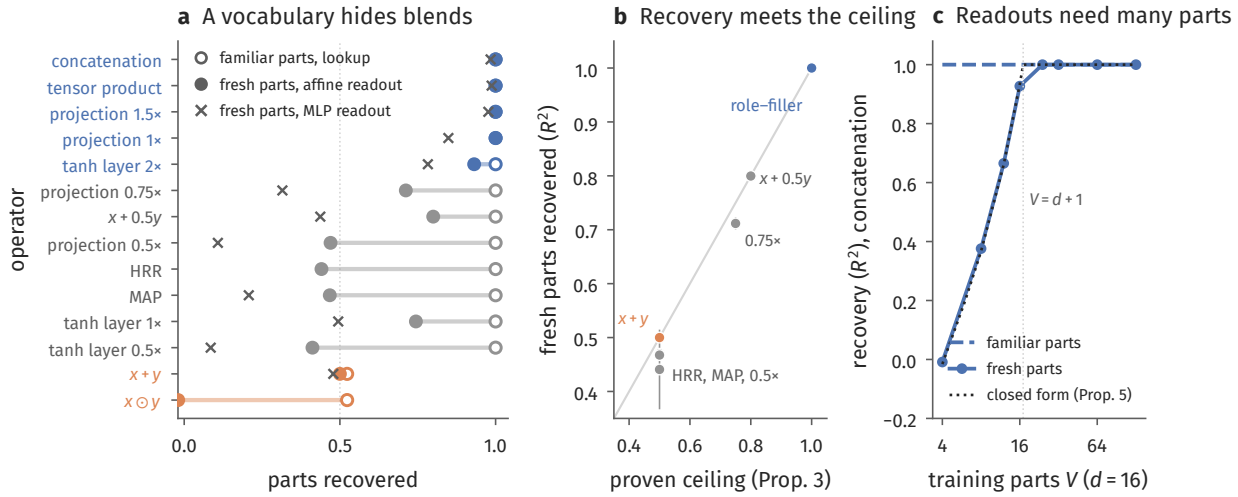

a A vocabulary hides blends
familiar parts, lookup
fresh parts, affine readout
fresh parts, MLP readout
concatenation
tensor product
projection 1.5×
projection 1×
tanh layer 2×
projection 0.75×
x + 0.5y
projection 0.5×
HRR
MAP
tanh layer 1×
tanh layer 0.5×
x + y
x ⊙ y
operator
0.0
0.5
1.0
parts recovered
b Recovery meets the ceiling
role–filler
x + 0.5y
0.75×
x + y
HRR, MAP, 0.5×
fresh parts recovered (R²)
proven ceiling (Prop. 3)
c Readouts need many parts
V = d + 1
familiar parts
fresh parts
closed form (Prop. 5)
recovery (R²), concatenation
training parts V (d = 16)